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PAPER_999: AGN F_U_Bi Merger with $S_{26}^{(3)}$

Abstract

We derive the buoyancy force F_{U_Bi} at the SMBH horizon of AGN merger systems using the 3rd-order Ramanujan summation $S_{26}^{(3)} = 0.095$. The recursive coefficient $R_n^{\{(d,k)\}} = \sum_{j=0}^{k-1} (-1)^j C(k-1,j)/(n+j)!$ provides alternating-sign corrections that reduce S_{26} from 0.57 (1st-order) to 0.095 (3rd-order).

1. Core Equation

$$F_{\{U,Bi\}^{\{AGN\}}} = \rho_{SCm} \cdot V \cdot S_{26}^{(3)2} \cdot |U_b|/(|U_g| + |U_b|)$$

For Centaurus A ($M_{BH} = 5.5 \times 10^7 M_\odot$, $a = 0.70$): $F_{U_Bi} = 5.47 \times 10^{35} \text{ m/s}^2$.

2. Ramanujan $S_{26}^{(3)}$

$$S_{26}^{(3)} = \sum_{n=1}^{26} (z^n/n^{26}) \cdot R_n^{\{(26,3)\}} = 9.500 \times 10^{-2}$$

Distinct from $S_{26}^{(1)} = 0.57$ due to higher-order alternating cancellation.

3. Implementation

File: fubi_agn_ns_mergers.py, class AGNFUBiMergerCalc. CP4 class #583.